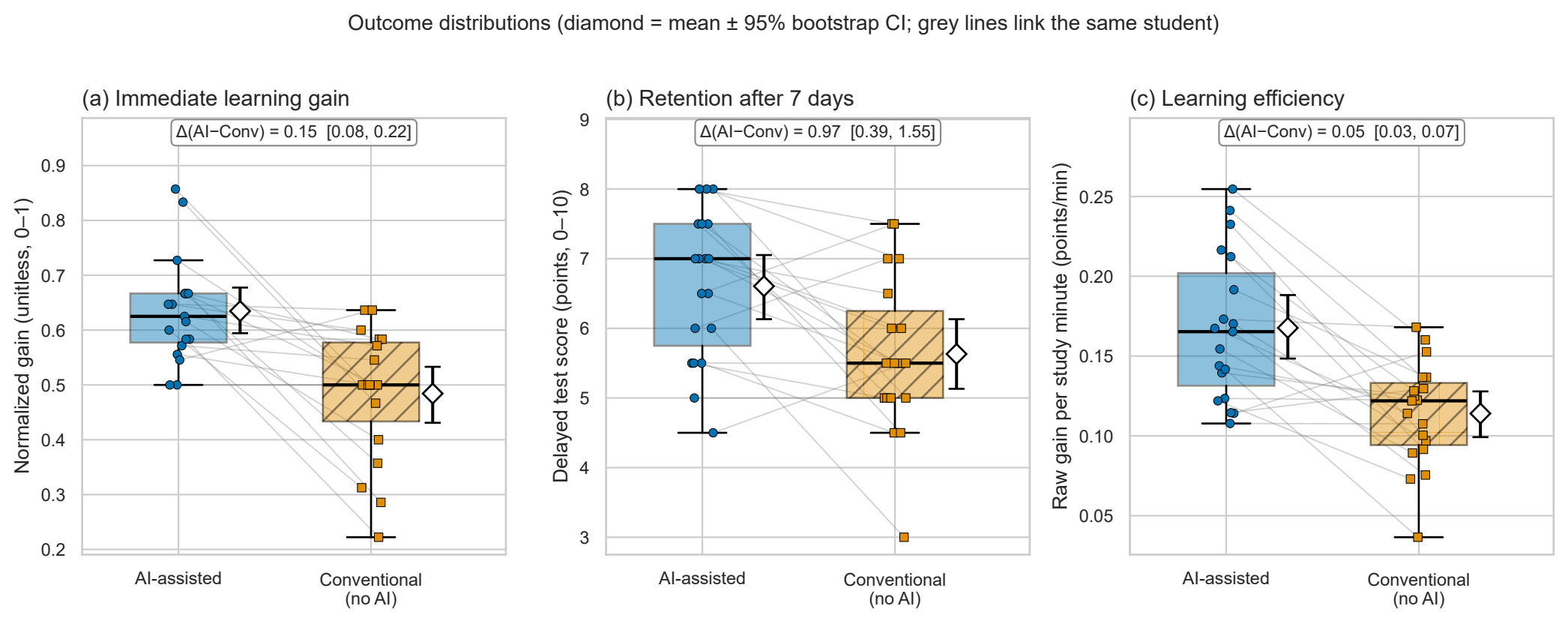
Outcome distributions (diamond = mean ± 95% bootstrap CI; grey lines link the same student)
(a) Immediate learning gain
Δ(AI−Conv) = 0.15 [0.08, 0.22]
Normalized gain (unitless, 0–1)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
AI-assisted
Conventional (no AI)
(b) Retention after 7 days
Δ(AI−Conv) = 0.97 [0.39, 1.55]
Delayed test score (points, 0–10)
9
8
7
6
5
4
3
AI-assisted
Conventional (no AI)
(c) Learning efficiency
Δ(AI−Conv) = 0.05 [0.03, 0.07]
Raw gain per study minute (points/min)
0.25
0.20
0.15
0.10
0.05
AI-assisted
Conventional (no AI)